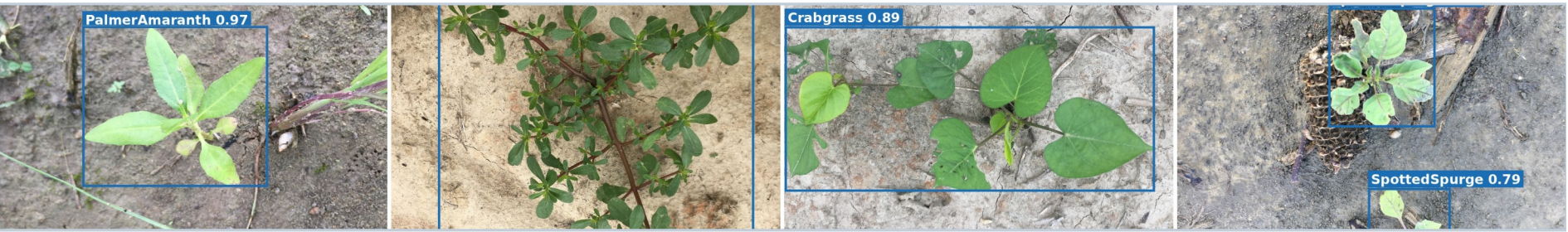


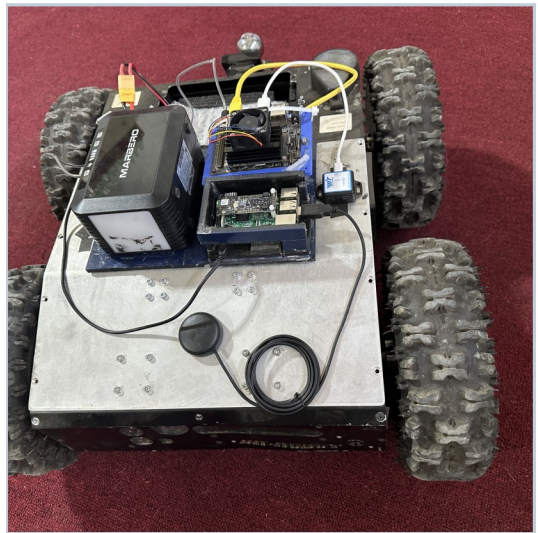
On one platform, agents collect their own data, review one another's work and train without an operator. The system ran for fifteen unattended rounds; this sheet reports how the agents coordinate and what the collection produced.

Weed Detection



The corpus exceeds 150,000 labelled images. Two agents operate on it unprompted: one harvests and audits imagery, the other trains a detector small enough for the robots.

Rover



A four-wheel skid-steer the width of a crop row carries a Jetson, a Raspberry Pi and its own battery. Camera, GPS and two IMUs stream live over one API key.

Laser Weed Control Robot



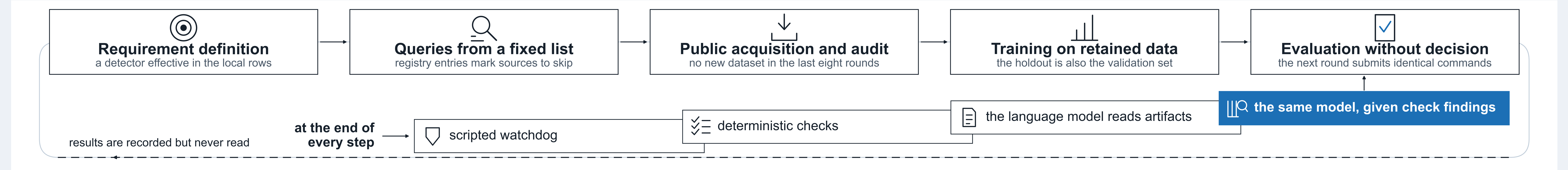
A YOLO detector localises each weed; two galvanometer lasers aim and fire, destroying it in place. No chemical is applied and nothing contacts the adjacent crop.

Humanoid and Quadruped



This project pairs a humanoid with a quadruped to record terrain that wheels do not reach. It adopts the Rover's ingest contract, so every session the robots open is governed.

Figure 1. Every project on the platform is created by the same procedure and registers its data under one ingest contract. The harvesting and training agents have so far run on one of them.



Agent Workflow Diagram

Figure 2. A requirement enters. The harvest issues fixed queries without gap analysis. Audited data train the model; the holdout is also the validation set and selects the reported epoch. Evaluation parses a metric; no decision follows. The reviewer runs after every step; the dashed edge is unimplemented.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

agentAI is one platform for every laboratory robot

A robot, drone or sensor rig is a project with its own keys, members and access-controlled dataset registry. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

Both the Rover and the laser robot transmit data

The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial measurement units (IMUs). The Laser Weed Control Robot carries two galvanometer lasers on independent channels, a downward camera over the work zone and a forward camera.

OPEN-WEIGHT MODELS REVIEW IN TIERS (CONT.)

Tiered review is invoked at the end of every step

The tiers are a scripted watchdog, twelve deterministic checks, a language model reading the artifacts with those checks withheld, then the same model supplied with them. On 149 frozen cases the scripted watchdog returned no decision.

Our own watchdog returned no decision on 149 dev cases

We froze 162 real failures out of this project's own engineering record and scored every kind of supervision against them. The watchdog the loop was running returned no decision on any of the 149 dev cases. A model reading the raw artifacts reaches 0.70 recall over 116 incidents.

A JUDGEMENT COUNTS ONLY WHERE IT RUNS (CONT.)

The corpus stopped growing at round three

The corpus stopped growing at round three: the same 24 datasets went in every time, and the apparent growth was one staging directory nobody cleared.

Each finished drive is diagnosed without anyone asking

Every finished drive is diagnosed without being asked -- route, noise, cross-sensor correlation, a timeline -- for 26 live robot sessions and 45 charts so far.

One hypothesis held, the other was ruled out

Both hypotheses were run as controls on round 15's exact data and seeds. The chain holds at 5.1 sigma; finishing the cosine schedule is worth -0.0058 at 1.3 sigma, inside the noise.

WHAT THE COLLECTION PRODUCED (CONT.)

One audited source in six clears the bar

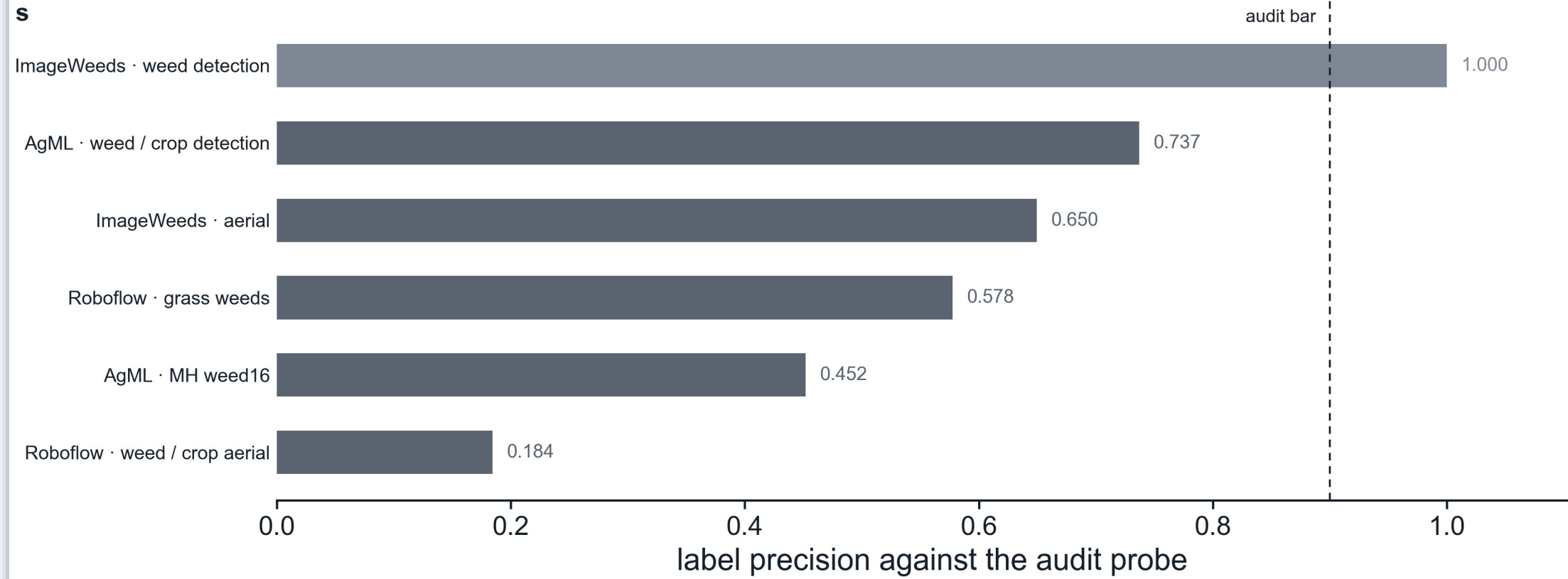
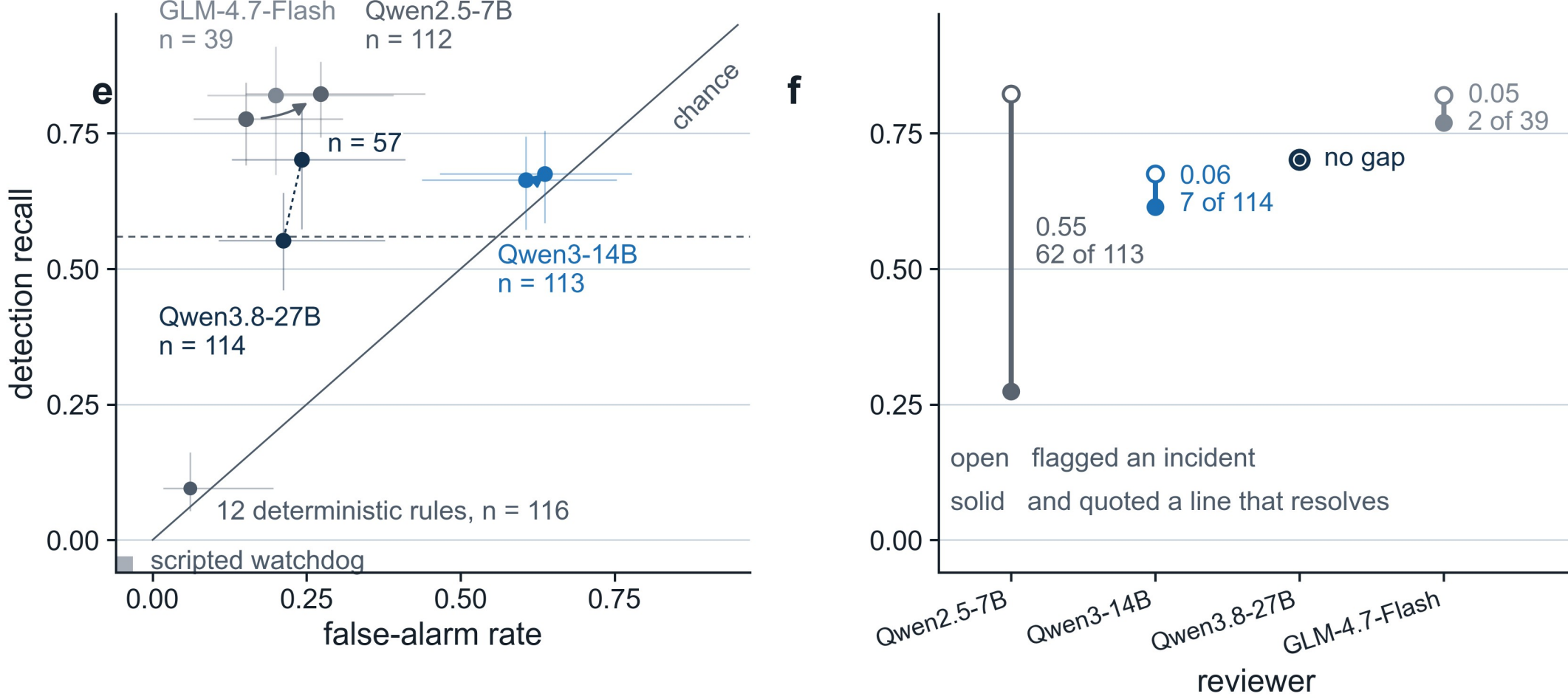


Figure 6. The six audited harvested sources. The same probe reads 1.000 on human-labelled CottonWeedDet12, so a source at 0.184 is the data and not the instrument. Only the source at the top clears the bar, and it supplies 3,208 of the 13,527 audited images.

OPEN-WEIGHT MODELS REVIEW IN TIERS

Grounded recall distinguishes the reviewers



Every reviewer is an open-weight model served from a self-hosted cluster. The higher tier adds grounding, not parameter count: supplying the deterministic tier's findings to the model yields +0.149 recall at 27 B and +0.012 at 14 B.

A JUDGEMENT COUNTS ONLY WHERE IT RUNS

A judgement counts only where it was produced

On the cluster		the answer is authoritative
plan the next step	qwen3.8 27B	
review in depth	glm-4.7-flash	
hard one-off reasoning	deepseek-v3 671B	
decide what to collect	gemma 4	
judge dataset quality	qwen2.5 7B	
label assistance	minicpm-v	
On the lab box		the answer returns marked a draft
turn intent into a project	qwen2.5 3B	
summarise a dataset	qwen2.5 3B	

Figure 4.

A router places each role on a model and a machine. Judgement produced on the cluster is returned authoritative; the same judgement produced on the laboratory box is returned marked a draft, so the rule travels with the answer.

Compounding was the premise of the unattended loop

One agent collects and one trains, so each round's model should exceed the last. Fifteen rounds ran unattended. The training corpus did not change after round three, yet the ledger and every round report recorded each round as new work.

WHAT THE COLLECTION PRODUCED

What the collection produced, at each gate

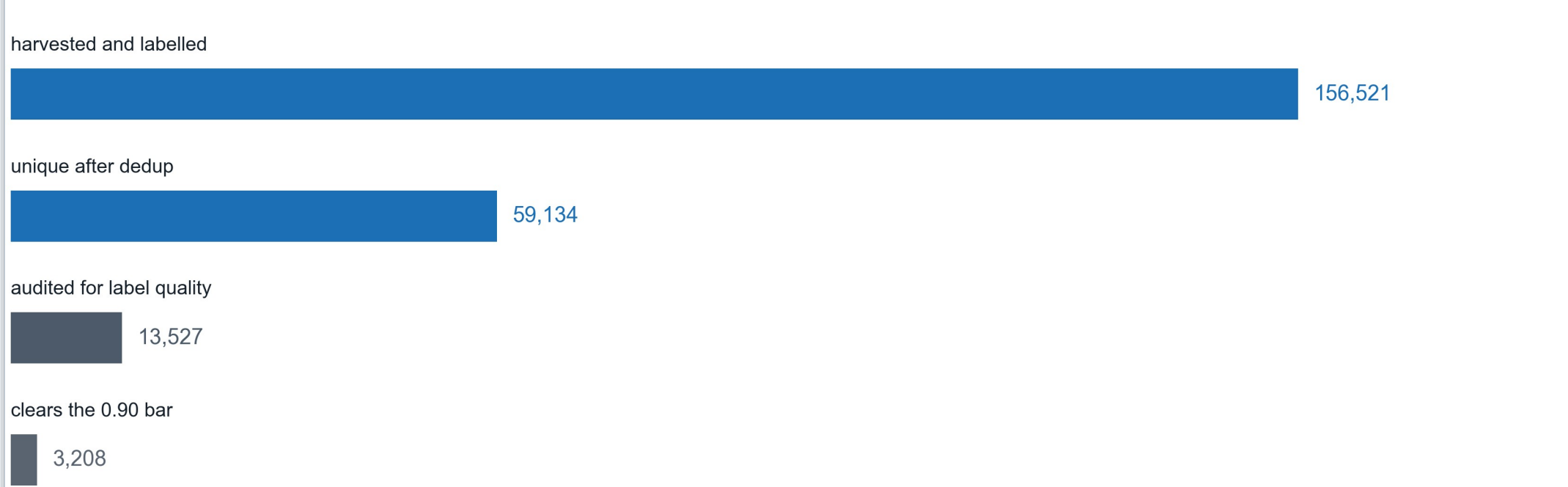


Figure 5. One of six audited harvested sources clears the label-precision bar. The audit probe reads 1.000 on human-labelled CottonWeedDet12, so the low scores are the data.

Automatic labelling wrote 156,521 images. Cross-dataset deduplication leaves 59,134 distinct ones, and of the 13,527 audited for label precision, 3,208 clear the bar.

Label precision is measured per source against a calibrated probe. The probe reads 1.000 on the human-labelled reference set, so a source reading 0.184 is the data and not the instrument.

The training configuration was not the constraint

Augmentation, replaying known species and completing labels were each tried and each left the same gap. What remains is the automatic labeller's boxes, and the audit above measures them.

Geometry rejects boxes the language model accepts

The rule discards boxes lying too far from the trusted data. Presented with a candidate box, Gemma 4 rejects none of the false boxes, so the criterion is DINOv2 geometry. Fifty-one datasets were removed.

The unit of curation must match the unit of noise

An automatic-labelling error is a single incorrect box in an otherwise acceptable image, so a dataset-level filter cannot detect it. The per-box tools are implemented but not re-measured. That remains an open question.

A language model's falsifiable hypothesis was false

Harvested boxes take their class from the source website. A language model proposed that as the cause, stated its falsifying condition first, and a human approved. Four runs falsified it. The cost was density: nine boxes per image against under two.